REVIEW OF FULL ADDER PERFORMANCE ANALYSIS USING KOGGE STONE ADDER AND MAGNETIC TUNNEL JUNCTION

Reshma Mary James
Dept. of Electronics and Communication Engineering
Saintgits College Of Engineering
jamesmaryreshma@gmail.com

Asst. Prof Ajith Ravindran
Dept. of Electronics and Communication Engineering
Saintgits College Of Engineering, Pathamuttom, Kerala
ajith.ravindran@saintgits.org

***Abstract*— Out of many adders like Ripple carry adder, carry look ahead adder, carry select adder and prefix adder, kogge stone adder is a high performance adder and also less affected by errors. Kogge stone adder is an advanced version of parallel prefix adders and also consists of less number of components thus delay is reduced. It consumes less power when compared with other adders, but it might consists of leakage current because of increase in stand-by power in the circuit. Leakage current is introduced in the circuit because of scaling of transistors. Thus VLSI designers found it difficult to reduce the parameters such as delay, power, area and leakage current simultaneously. To overcome this disadvantage, an MTJ(Magnetic Tunnel Juction) is introduced in the CMOS circuits. By the use of MTJs in the circuit the leakage current is reduced and zero standby power is obtained. A comparison analysis is done between a hybrid 4-bit MTJ/CMOS based full adder and 4-bit Kogge stone adder.**

Keywords—Kogge stone adders, Magnetic tunnel junction, Magnetic Full adder

I. INTRODUCTION

VLSI designers are struggling to meet the requirements of the parameters such as delay, power, area and speed. After the use of many methods and techniques scaling was introduced assuming that it would overcome the disadvantages which was faced earlier. Scaling leads to shrinking of transistors which reduces the size of transistors and also reduces power dissipation of the circuit. Earlier moore's law was used instead of scaling. scaling was introduced to increase the current density and speed of the circuit[22].

Scaling reduces the area and power but it leads to many issues such as Sub threshold leakage, leakage current, Drain induced barrier lowering, Reverse short channel effect, Early voltage, output resistance , Linear proximity effects, Non-linear proximity effects and Gate induced drain leakage. Since the transistor size was reduced the circuit becomes more complicated since many transistors was included in an single chip. Due to scaling gate direct tunnelling current is increased, when ever a negative voltage is driven by gate it leads to gate induced barrier leakage. Generally gate induced barrier lowering is occurred in SRAMs[23].

Scaling reduces the reliability and also effects the speed of the circuit, thus threshold voltage and supply voltage are only reduced to a particular limit. As the supply voltage goes on reducing it increases the electric field strength this leads to degradation of carrier mobility and may damage the device. Due to short channel effects the gate voltage is decreased below the threshold voltage thus causing drastic effects such as punch through, threshold voltage roll-off, and drain induced barrier lowering [20].

Multichip devices are heterogeneous and therefore it consists of different circuits with multiple power supply. Scaling of multiple power supply leads to compatibility problems. As the threshold voltage is reduced it causes the sub-threshold voltage to increase which decreases the speed and power efficiency. The cost of the chip can be reduced due to scaling, since the size is reduced to a large extend which might cause errors during operation hence more improved test methods are used. The test methods used to analyze the operation of the circuit may be costly. Therefore scaling must be done in such that it must not affect circuit performance[21].

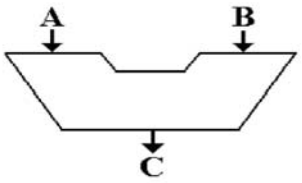


Fig. 1.1. ALU

ALU is an basic adder that was initially introduced for performing all the arithmetic and logic operations as in Fig. 1.1. shown above. But ALU limits the performance and speed of the circuit.

By using static CMOS circuits like Inverting and non-Inverting CMOS, Footed and Footless Domino logic

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adder circuits can be built. Adder circuits consists of propagate, generate bits, which can be further subdivided into black, grey cells. Black cells are necessary when output driving upper inputs of other PG cells. Similarly, grey cells are necessary when output driving lower inputs. The Logic effort and Parasitic delay of many logic circuits and adders are evaluated as in [11]. The Logic effort can be defined as ratio of input capacitance of gate to input capacitance of invertor. The parasitic delay is calculated by the transistor width of output node[11].

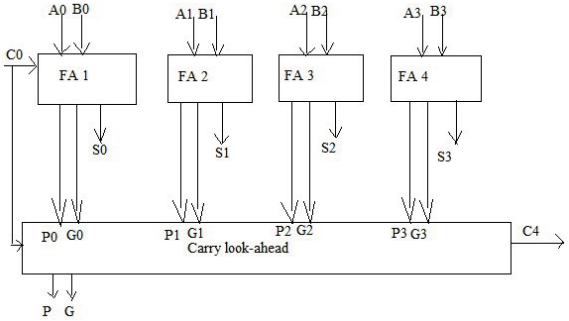


Fig. 1.2. Carry look-ahead adder

In Carry look ahead adder carry, generate, and propagate bits can be calculated without any ripple effect as shown in Fig 1.2. Carry look ahead (CLA) adder has to wait for a long time since it has to calculate previous carry bits therefore Carry select adder is better than CLA[3].

$$P_0=a_0+b_0 \quad (1.1)$$

$$G_0=a_0b_0 \quad (1.2)$$

$$C1=G_0+P_0C_0 \quad (1.3)$$

In carry select adder multiple carries are independently produced and one carry is chosen among them, thus our propagation delay can be eliminated. Carry select adder is having an simple architecture and is faster than earlier two adders as in the Fig. 1.3 shown below[2].

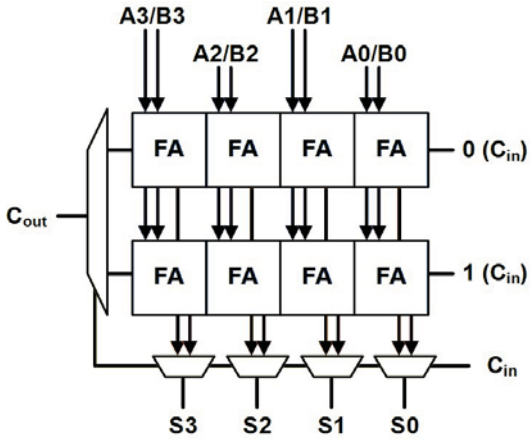


Fig. 1.3. Carry select adder

Prefix adders pre-calculate the carry signals so that propagation delay can be reduced which was a problem in the earlier circuits. Parallel prefix adders also consists of three unit such as Pre-processing, Prefix computation and Post-processing. In pre-processing unit generate, propagate signals are produced.

$$G_i=a_ib_i \quad (1.4)$$

$$P_i=a_i \; xor \; b_i \quad (1.5)$$

In pre-computation unit depends on the group of carry propagate, generate signals

$$(G,P)_{[i:k]} = (G,P)_{[i:j]} * (G,P)_{[j-1:k]} \quad (1.6)$$

Finally, in post-processing unit sum and carry is produced for each operand bits.

$$C_i = G_{[i:0]} \quad (1.7)$$

$$S_i = P_i \; xor \; C_{i-1} \quad (1.8)$$

Now a days most commonly used adder is Kogge stone adder which is presently discussed in this paper. Kogge stone adder is the modified version of prefix adder, which has less propagation delay hence fastest[12].

Leakage current is caused due to the path created between supply voltage and ground. An MTJ is placed between supply voltage and ground, thus problem of leakage current is eliminated[1]. By varying direction of magnetic moment, we can alter the electron flow this mechanism is used in magnetic tunnel junction[7]. An MTJ consists of two electrodes made up of ferromagnetic materials which is separated by an insulating layer. The magnetization of electrodes can be changed by applying an electric current either in spin up or spin down direction. The magnetization of the two electrodes happen either in anti-parallel or parallel direction. If two electrodes are parallel means the magnetic directions are same otherwise the magnetic directions are different[8]. The resistance offered by an MTJ when the electrodes are in both anti-parallel and parallel direction are R_P and R_{AP}. The difference between the resistances offered after the eletrodes are magnetized in both parallel and anti-parallel are called Tunnel magneto resistance (TMR)[9].

$$TMR=\frac{R_{AP}-R_P}{R_P}\times 100 \quad (1.9)$$

One of the most important application of magnetic tunnel junction is it acts as an non-volatile memory. By varying the magnetic field there is a change in both junction resistance and junction capacitance. At higher frequencies MTJs performance can be connected to the dielectric properties, this effect is known as Tunnel magneto capacitance (TMC) effect[10]. Based on MTJ a magnetic flip flop with non-volatile memory was introduced, which constructs the circuit with less number of errors and it can be emerged in SOC design to produce low power. The soft errors such as single event upsets can be prevented by using MTJs[5].

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II. KOGGE STONE ADDER

Kogge stone adder was considered as the modified version of prefix adders which focuses on time and also assures higher performance. Kogge stone adder is made up of three units that is pre-processing, carry look ahead network and post-processing. Kogge stone adder is one of the speediest circuit and also it can tolerate to any type of defect.[18]

In pre-processing stage, the propagate and generate bits are obtained by the following equation:

$$P_i = A_i \; xor \; B_i \tag{2.1}$$

$$G_i = A_i \; and \; B_i \tag{2.2}$$

In carry look ahead network, it computes corresponding carry to each bit. Group of carry and propagate bit are used as intermediate stage which are obtained by the equation:

$$P_{i:j} = P_{i:k+1} \; and \; P_{k:j} \tag{2.3}$$

$$G_{i:j} = G_{i:k+1} \; or \; (P_{i:k+1} \; and \; G_{k:j}) \tag{2.4}$$

In post-processing stage, finally the sum bit is evaluated using the equation:

$$S_i = P_i \; xor \; C_{i-1} \tag{2.5}$$

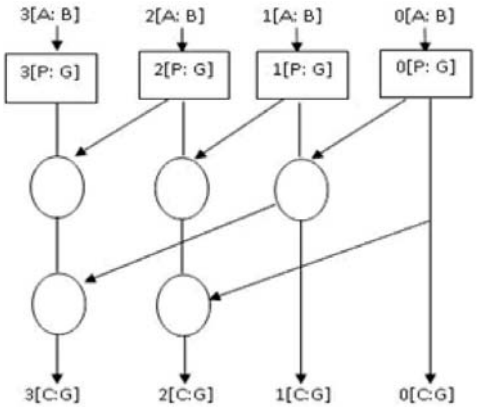


Fig. 2.1. Kogge stone adder(Vishal Galphat),2015

A. 4-bit Kogge stone adder

The Fig 2.2 shows the gate level schematic of 4-bit Kogge stone adder. A 4-bit Kogge stone adder will reduce the delay when compared to other adders, thus it is one of the most fastest adders. It is the combination of half adders, black cells, grey cells. Grey cell will produce the carry for each corresponding operand. Black cell will produce the propagate and generate signals, thus in the intermediate stage both black cells and grey cells are combined to produce the output sum. In Kogge stone adder it might consists of leakage current it would lead to large power consumption to overcome this problem an MTJ is integrated in the adder circuit.

Grey cell can be written as:

$$C_0=p_0cin+g_0 \tag{2.6}$$

Black cell can be written as:

$$P=p_0p_1 \tag{2.7}$$

$$G=p_1g_0+g_1 \tag{2.8}$$

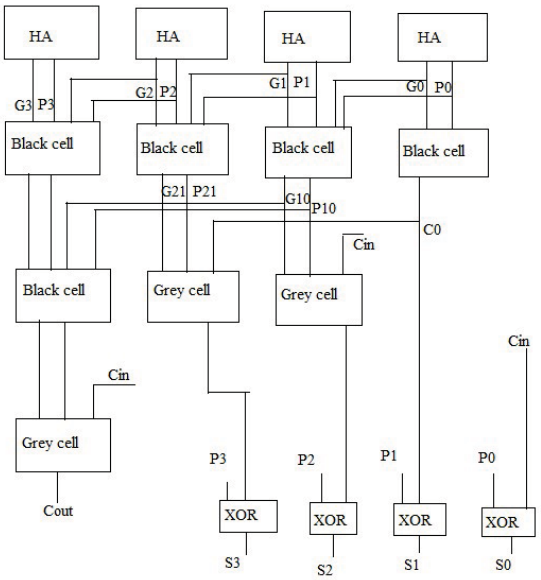


Fig 2.2 4-bit Kogge stone adder

B. Application of Kogge stone adder

A high speed Kogge stone adder was introduced using mux, thus it can be preferred for any processor based application[19]. kogge stone adders can also be used in high speed applications. By using Kogge stone adders an Vedic multiplier can be implemented, thus this fastest adder is widely used in industries[17]. Braun multiplier is generally implemented using ripple carry adder, AND gate and carry save adder. To improve the performance ripple carry adder was replaced with Kogge stone adder, thus delay is reduced.

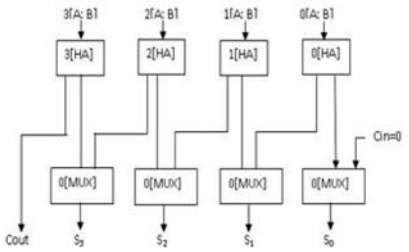


Fig 2.3 Kogge stone adder using MUX(Vishal Galphat),2015

III. MAGNETIC FULL ADDER

MTJ was greatly used in circuits because of its non-volatility and also it is of small size and compatible

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with other CMOS technologies. In the absence of power supply MTJs have the capability to store the data, hence it is known as magnetic storage device. When the input current given to MTJ is varied to a positive direction, then MTJs resistance is transformed from parallel state to antiparallel state. when the input current offered to MTJ is decreased to zero, then MTJs resistance is not changed indicating MTJs non-volatile memory property. Similarly, when input current is varied from zero to a negative value, then MTJs resistance is altered from antiparallel to parallel state. Hence data is not lost in the case of MTJ even if value is decreased to zero from negative value, therefore data remains in MTJ even when there is no power[26].

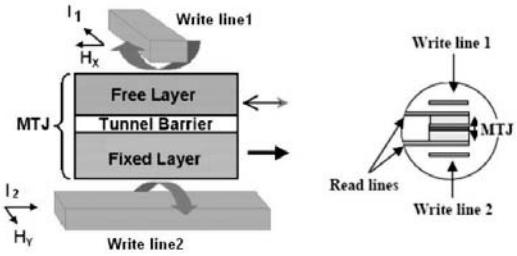


Fig 3.1 Schematic structure of MTJ(Noboru Sakimura,2008)

MTJ can be used instead of storage inverters in the SRAM cell. MTJs are driven by current in place of voltage. An MTJ register was constructed which would act as an important component of circuit. As in Fig 3.1 shown above magnetic field is generated by adding two write lines such that data in the MTJs can be changed depending upon the currents flowing through two write lines. By applying a magnetic field at the junction of the MTJ the data can be written in the circuit[13]. As in Fig 3.2 shown below represents the D flip flop which has been constructed using MTJ.

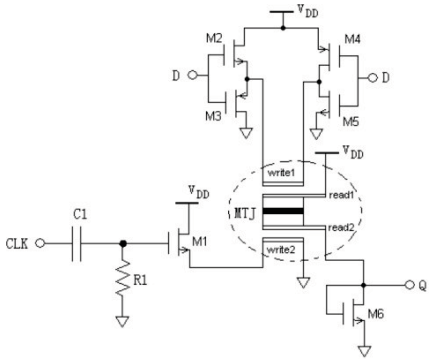


Fig 3.2 D flip-flop using MTJ(Noboru Sakimura,2008)

Input signals CLK and D should be transformed to current signals. CLK and D are applied to write lines 2 and 1, thus current signals I2 and I1 are produced. The current flowing through write line 1 decides the state which has to be written in an MTJ register. I1 is in parallel or antiparallel direction with respect to MTJs free layer depending upon input signal D= 1 or 0.

An MTJ consists of fixed layer (FL) and pinned layer (PL) where fixed layer takes less switching time, pinned layer acts as a spin polarizer. Magnetic tunnel junction consists of two polarizations such as perpendicular and longitudinal polarization depending upon origin of free layer magnetization. Due to low switching current density perpendicular devices are mostly used. Current that is induced in the junction is equivalent to magnetization generated between the PL and FL layers and also depend on temperature[15].

Current density required for originating a magnetic excitations in both longitudinal and perpendicular directions are:

$$J_C^{LONG} = \alpha \frac{2e}{\hbar} \frac{M_s.t}{g(\theta=0)} (H_k + 2\pi M_s) \quad (3.1)$$

$$J_C^{PERP} = \frac{2e}{\hbar} \frac{M_s.t}{g(\theta=\frac{\pi}{2})} \frac{H_k}{2} \quad (3.2)$$

Where J_C^{LONG} and J_C^{PERP} are current densities of both longitudinal and perpendicular polarization. H_k is in-plane anisotropy originated from crystalline.

An polarizing layer is added to STT-MTJ, thus current is spin polarized. When a critical switching current is applied on MTJ, STT-MTJ reaches critical current density at a specific time period hence alters the state of MTJ. Circuit has been simplified since additional write lines was eliminated which was used to handle the device. It discusses about the TMR ratio and its relation with bias voltage. By using this model we can obtain error-free results for parameters of MTJ and also fast simulation of CMOS circuits using MTJ[25].

Electrons consists of both wave nature and particle nature. An electron tunnel's through dielectric layer because of its wave nature due to the change in magnetization of the two ferromagnetic layer. According to Simmon's model we are analyzing the tunneling transport of electrons across the junction. TMR ratio was modeled by Julliere's and Slonczewski model relying on factors such as barrier height and bias voltage.MTJs tunnel magneto resistance and antiparallel resistance decreases because tunneling probability decreases across the insulator with increase in barrier height which further rely on junction capacitance. For limited bias voltage julliere's model was used to produce accurate TMR. Temperature variations effects the circuits performance by changing the magnetic and electrical properties of the MTJ[14].

Pseudo spin value (PSV) MTJs are having higher TMR at an annealing temperature exceeds 450°C, when compared to spin value (SV) MTJs. This large difference was produced because of lack of spreading of Mn onto Mgo barrier layer at higher annealing temperature. A study was made so that we could determine TMR ratio dependence on annealing temperature for various thickness and compositions of an PSV MTJs. TMR ratio is increased upto 450°C and then it decreases beyond 500°C. Analysis is done depending upon how TMR ratio is decreased when annealing temperature rises. This happened due to scattering of tantalum (Ta), when temperature increases. Therefore by reducing the

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spreading of Ta at higher temperature, TMR ratio increases[6].

A. 4-bit MTJ based hybrid fulladder

The MTJ/CMOS based full adder consists of a write circuit, sense amplifier and a logic tree. The MTJ cells present in each path are error tolerant, hence any particle strikes does not effect the circuit. The leakage current is also eliminated. Sense amplifier performs read operation. Similarly if there is any change in data is induced by the write circuit. Logic tree is a combinational logic circuit that indicates and perform the function of the full adder circuit. The circuit as shown in the Fig 3.3 consists of four MTJ cells[1].From the Fig 3.3 the sum output is constructed by using the equations 3.3 and 3.4. Similarly the carry equation was constructed by using the equations 3.5 and 3.6.

$$Sum = ABC + A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} \quad (3.3)$$

$$\overline{Sum} = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C + AB\bar{C} \quad (3.4)$$

$$cout = AB + BC + CA \quad (3.5)$$

$$\overline{cout} = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{C}\bar{A} \quad (3.6)$$

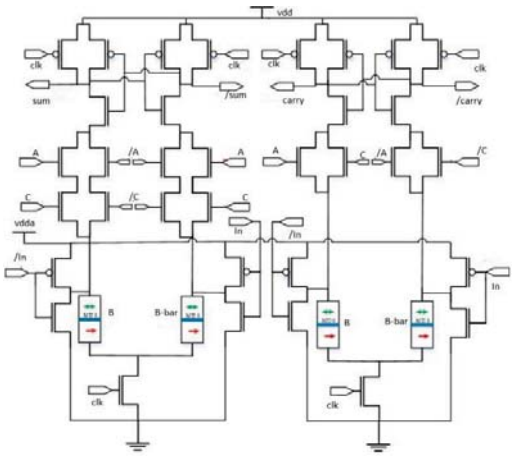


Fig 3.3 Magnetic full adder circuit(Surya Kiran Akkaladevi,2015)

The above circuit has three inputs A, B AND C. The inputs A and C are different then AC and $\overline{AC}$ must have same resistance otherwise they have different resistance, thus corresponding outputs sum and carry is obtained.

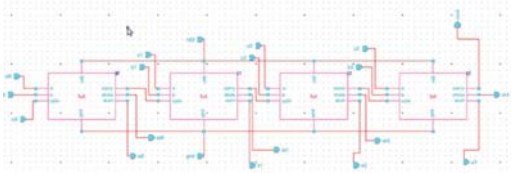


Fig 3.4 Schematic of 4 bit MFA

The Fig 3.4 indicates a 4 bit MTJ/CMOS based full adder. Each full adder consists of four MTJ cells, therefore a total of 8 MTJ cells.The power consumed by the 4 bit MTJ/CMOS based full adder is low, hence MTJ is used in other CMOS circuits and also less sensitive to errors. MTJs output is obtained and verified using Verilog A language and by using this symbol a 4-bit MTJ based full adder circuit is obtained.

B. Applications of MTJ

MTJs acts as an non-volatile memory and also reduce the stand by power induced in the circuit. STT-MTJ is used as an reprogrammable memory which performs Boolean logic operations such as NAND, AND, NOR and OR, this operation is also known as material implementation. Whenever a current I_{IMP} is induced to the I_{IMP} gate depending the initial conditions it changes the orientations of initial and final MTJ. The output of the IMP operation is produced at the final MTJ. The 1T/MTJs are replaced instead of MTJs as shown in the Fig 3.5, Since it function as a non-volatile magnetic circuit[16]. MTJ has a lot of applications like high frequency detector and magnetic field sensor.

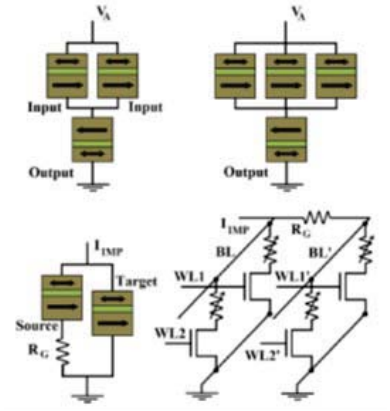


Fig 3.5 Reprogrammable two input structure using MTJs(Hiwa Mahmoudi),2013

Other application of an MTJ is inducing MTJs in MRAMs as in Fig 3.6 shown below. MRAMs includes MTJs in which base electrodes is used as write lines. MTJ is placed on it so that it can improve the orientation of magnetic field between the ferromagnetic layers. It reduces the current when compared to conventional MRAMs[5].

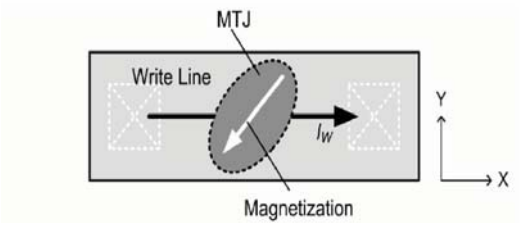


Fig 3.6 MRAM technology using MTJ(Noboru Sakimura),2008

When compared to conventional MRAMs the operation carried out by the proposed MRAMs are very fast. The sensing current and the I_w path are different so disturbances during read operation can be avoided.

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IV. Result and Discussions

A 4-bit full adders was created using MTJ such that delay is minimized to a great extent. The TMR ratio of the MTJ is made higher such that it increases the circuit performance. Verilog-A language is preferred to implement the MTJ, such that a symbol was created in Cadence IC 6.1.6 which illustrates the change of state of MTJ from anti-parallel to parallel and vice versa. Temperature variations of circuit is also taken into consideration. MTJ is placed in between the VDD and GND such that it minimizes the leakage current which was one of the main disadvantages of conventional CMOS. VCC=1v was chosen for 4-bit MTJ/CMOS based hybrid adders. Carry-out(cout) and input(a0) of MTJ/CMOS based hydrid adders is chosen to calculate delay. Power was also calculated such that we can ensure which one has higher performance. Simulation is done in spectre as in the Fig 4.1. shown below.

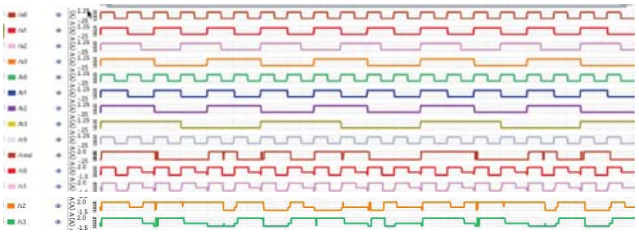


Fig 4.1 Output of 4-bit full adders

VDD=1.8v was chosen for Kogge stone adder, Carry-out(cout) and input (a0) was chosen to calculate the delay. The power is also calculated ,so that a comparison analysis is made.

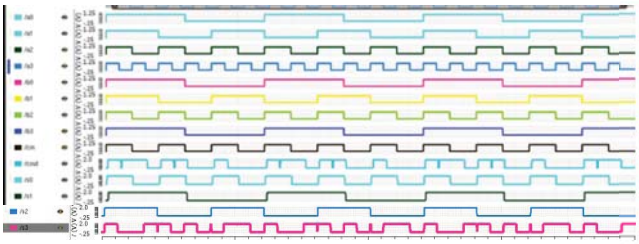


Fig 4.2 Output of 4-bit Kogge stone adder

The above Fig 4.2 is the output of 4-bit Kogge stone adder. A comparative analysis between MTJ and hybrid adders is done such that we can analyse which has better performance.

The comparison between 4-bit full adders, 4-bit MTJ/CMOS based full adders and 4-bit Kogge stone adder has been analyzed as in the table 5.1 shown below. In table 5.1 parameters including delay, power, area, and PDP(power delay product) has been evaluated. Delay and power was measured using cadence tool, by using these two parameters PDP has been evaluated and area was analyzed by taking the transistor count from the schematic created in the cadence tool.

Table 5.1 Comparison Analysis

	Delay	**Power**	**Area** (Transistor count)	**PDP**
4-bit Full adder	5.093ns	3.40mw	144	17.31wps
4-bit MTJ/CMOS based full adders	2.32ns	4.201mw	112	9.74wps
4-bit Kogge stone adder	14.6ns	1.146mw	320	16.73wps

V. Conclusion

By comparing the results of 4-bit Kogge stone adder and 4-bit full adders using MTJ for there corresponding inputs (a0a1 a2 a3=1011 & b0b1b2b3=1011 & c0=1),we can say that the PDP of an MTJ based full adder is reduced to 8% and 2% when compared to conventional full adders and Kogge stone adder. There is an increase in delay of Kogge stone adder when the power is reduced, thus in the case of MTJ it maintains a trade-off between power and delay, hence introduction of MTJ is an better option.

References

[1] Ramin Rajaei, Sina Bakhtavari Mamaghani , " Ultra-Low Power, Highly Reliable, and Nonvolatile **Hybrid MTJ/CMOS Based Full-Adder for Future VLSI Design," IEEE Transactions on Device and Materials Reliability ,2016.**

[2] B.Tapasvi, K.Bala Sinduri, "Implementation of 64-Bit Kogge Stone CarrySelect Adder with ZFC for Efficient AreaImplementation of 64-Bit Kogge Stone Carry Select Adder with ZFC for Efficient Area", IEEE International Conference on Electrical, Computer and Communication Technologies,2015

[3] Aritra Mitra, Amit Bakshi, Bhavesh Sharma, Nilesh Didwania, "Design of a High Speed Adder", International Journal of Scientific & Engineering Research, Volume 6, Issue 4, April-2015

[4] S Chopra, S Subramaniam, "A review on challenges for MOSFET scaling" Int. J. Innovative Science, Volume 2, Issue 4, April 2015

[5] Sakimura, Noboru, et al. "Nonvolatile magnetic flip-flop for standby-power-free SoCs." Solid-State Circuits", IEEE Journal of 44.8 (2009): 2244-2250

[6] S. Ikeda, J. Hayakawa, Y. Ashizawa, Y. Lee, K. Miura, H. Hasegawa, M. Tsunoda, F. Matsukura, and H. Ohno, "Tunnel magnetoresistance of 604% at 300 k by suppression of ta diffusion in cofeb/mgo/cofeb pseudo-spin-valves annealed at high temperature," Applied Physics Letters 93, 082508–082508 (2008)

[7] Jian-Gang (Jimmy) Zhu, Chando Park, "Magnetic tunnel junctions, " Data Storage Systems Center, Department of Electrical and Computer Engineering,

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Carnegie Mellon University, Pittsburgh, Volume 9, Issue 11, pp. 36–45

[8] E. Deng, Y. Wang, Z. Wang, J. O. Klein, B. Dieny, G. Prenat, W. Zhao, “Robust Magnetic Full-Adder with Voltage Sensing 2T/2MTJ Cell,” IEEE/ACM International Conference Symposium on Nanoscale Architectures, 2015.

[9] W. Zhao, E. Belhaire, C. Chappert, P. Mazoyer, "Spin Transfer Torque (STT)-MRAM based Run Time Reconfiguration FPGA circuit," ACM Transactions on Embedded Computing Systems, 2009

[10] Yin-Ming Chang, Kai-Shin Li, Hongbo Huang, Mean-Jue Tung, Shi-Yuan Tong, and Minn-Tsong Lin, “Extraction of the tunnel magneto capacitance with two-terminal measurements, ” Journal of Applied Physics 107, 093904 (2010)

[11] D. Harris, and I. Sutherland. ‘‘Logical effort of carry propagate adders.’’ Conference Record of the Thirty-Seventh Asilomar Conference on Signals, Systems and Computers, vol. 1, pp. 873–78, 2003.

[12] Bart R. Zeydel, Dursun Baran and Vojin G. Oklobdzija, “Energy-Efficient Design Methodologies: High-Performance VLSI Adders”, IEEE Journal of solid-state circuit, Vol. 45, No. 6, June 2010

[13] Yanfeng Jiang ; Jiaxin Ju ; Xiaobo Zhang ; Bing Yang, “Design and implementation of MTJ-based register” IEEE Institute of Electrical and Electronics Engineers, pp. 303 – 306, 20 August 2009

[14] W Guo, G Prenat, V Javerliac, M El Baraji, N De Mestier, C Baraduc, B Di_eny, “SPICE modelling of magnetic tunnel junctions written by spin-transfer torque”, Journal of Physics D: Applied Physics, 43 (21), pp.215001,2010

[15] A. Mejdoubi, G. Prenat, and B. Dieny, “A compact model of precessional spin-transfer switching for MTJ with a perpendicular polarizer”, PROC. 28th International Conference On Microelectronics, Serbia, pp.13-16 May, 2012

[16] Hiwa Mahmoudi, Thomas Windbacher, Viktor Sverdlov, and Siegfried Selberherr, “Design and Applications of Magnetic Tunnel Junction Based Logic Circuits”, Institute for Microelectronics, pp.157-160,2013

[17] Anjana, R., B. Abishna, M. S. Harshitha, E. Abhishek, V. Ravichandra, and M. S. Suma. "Implementation of vedic multiplier using Kogge-stone adder." In IEEE International Conference on, Embedded Systems (ICES), 2014 pp. 28-31, 2014.

[18] Swaroop Ghosh, Patrick Ndai, Kaushik Roy. "A Novel Low Overhead Fault Tolerant Kogge-Stone Adder Using Adaptive Clocking". DATE 2008

[19] Vishal Galphat, Nitin Lonbale ECE Deptt, SBITM, Betul, M.P., “ Design the High Speed Kogge-Stone Adder by Using MUX”, Int. Journal of Engineering Research and Applications, Vol. 5, Issue 8, pp.58-60, (Part - 2) August 2015

[20] Saaty, T.L, “A scaling method for priorities in hierarchical structures”, J. Math. Psychol. 9, pp. 234-281, 1977

[21] H. Iwai, "CMOS Scaling for sub-90 nm to sub-l0 nm", Proc. of the 17th IntI. Conference on VLSI Design (VLSID04), pp. 30-35, 2004

[22] E.J Novak, “Maintaining the benefits of CMOS scaling when scaling bogs down”, IBM J.Res & Dev., Vol. 46, NO. 2/3, 2002

[23] T. C. Chen, "Overcoming Research Challenges for CMOS Scaling: Industry Directions", Proc. of 8th Intl. Conference on Solid-State and Integrated Circuit Technology, 2006 (ICSICT'06), pp. 4-7, 2006.

[24] Madhu Thakur and Javed Ashraf ,“Design of Braun Multiplier with Kogge-Stone Adder & It’s Implementation on FPGA”, International Journal of Scientific& Engineering Research, Vol. 3, No. 10,pp. 03-06, 2012.

[25] R. Garg, D. Kumar, N. Jindal, N. Negi, and C. Ahuja, “Behavioural model of spin torque transfer magnetic tunnel junction, using Verilog-A,” International Journal of Advanced Scientific and Technical Research, vol. 1, no. 6, 2012.

[26] Chandramauleshwar Roy, Arundhati Bhattacharya, A. Islam, “Modeling of MTJ and its validation using Nano-scale MRAM bit cell,” Journal of Engineering Science and Technology Vol. 12, No. 6, pp.1525 – 1540,2017.

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